

Office Workers' Objectively Measured Sedentary Behavior and Physical Activity During and Outside Working Hours

Stacy A. Clemes, PhD, Sophie E. O'Connell, PhD, and Charlotte L. Edwardson, PhD

Objective: To examine objectively determined sedentary behavior and physical activity (PA) during and outside working hours in full-time office workers. **Methods:** A total of 170 participants wore an ActiGraph GT1M accelerometer for 7 days. Time spent sedentary (<100 counts/min), in light-intensity PA (100 to 1951 counts/min), and moderate-to-vigorous PA (≥ 1952 counts/min) was calculated for workdays (including working hours and nonworking hours) and nonworkdays. **Results:** Participants accumulated significantly higher levels of sedentary behavior (68% vs 60%) and lower levels of light-intensity activity (28% vs 36%) on workdays in comparison with nonworkdays. Up to 71% of working hours were spent sedentary. Individuals who were most sedentary at work were also more sedentary outside work. **Conclusions:** Those who are most sedentary at work do not compensate by increasing their PA or reducing their sedentary time outside work. Occupational interventions should address workplace and leisure-time sedentary behavior.

Sedentary behavior, defined as “any waking behaviour characterised by an energy expenditure ≤ 1.5 METs [metabolic equivalents] while in a sitting or reclining posture” (page 540),¹ is an independent risk factor for a number of adverse health outcomes. For example, greater sitting time (the terms “sitting” and “sedentary behavior” are used interchangeably herein) has been associated with an increased risk of overweight, obesity and weight gain,^{2,3} cancer,^{4,5} type 2 diabetes and the metabolic syndrome,^{2,6} all-cause mortality, and cardiovascular disease mortality,^{7,8} independent of moderate-to-vigorous physical activity (MVPA). There is a growing consensus that sedentary behavior represents a unique aspect of human behavior and that it should not be viewed simply as the absence of physical activity.^{7,9}

Adults typically spend time sitting in three domains: the workplace, during leisure, and for transport.¹⁰ Economic advances and industrial innovation have resulted in large numbers of people employed within sedentary occupations, and data from Australian workers have shown that half of their total daily self-reported sitting time takes place at work.^{11,12} Accelerometer data from Australian office workers have shown that between 66% and 82% of their working day is spent sedentary.^{13–15} Of concern, it has been observed in some studies that those who are sedentary for a large proportion of their working day do not compensate by increasing their physical activity levels and/or reducing their sedentary behavior during leisure time.^{12,14,16,17}

Our understanding of the prevalence of sedentary behavior in UK adults is currently limited, and has largely been restricted to the study of leisure-time screen-based sedentary behaviors¹⁸ or to specific occupational groups, such as postal workers.¹⁷ It is important to measure sedentary behavior and physical activity across a range of domains, particularly the workplace, if we are to truly understand patterns and determinants of these behaviors in adults, to inform behavior change interventions.¹⁹ To date, limited research has examined objectively measured sedentary time during and outside working hours.^{13,14} Increasing our understanding of the potential impact of sedentary behavior during work, on sedentary behavior and physical activity outside work, has been highlighted as a research priority.¹³ The aim of this study therefore was to examine objectively determined sedentary behavior and physical activity occurring during and outside working hours in a sample of full-time office workers from the United Kingdom. A secondary aim was to build on our understanding of the links between sedentary behavior accumulated during and outside working hours by investigating whether those who are sedentary for a large proportion of their working hours compensate by decreasing their sedentary behavior, or increasing their physical activity, during nonworking hours.

METHODS

Participants

A convenience sample of 210 office workers was recruited from Loughborough University and local businesses within the East Midlands region of the United Kingdom. The study inclusion criteria ensured that all participants were aged between 18 and 65 years and in full-time office-based work. Responses on a health screen questionnaire completed at the outset confirmed that participants were all in good general health, with no reported physical illnesses or disabilities that may affect their normal daily routine. The sample consisted of individuals employed within administrative roles, and all participants described themselves as having a predominately sedentary occupation. The standard working hours of the organizations involved were 9 AM to 5 PM on Mondays to Fridays. The study received ethical approval from the Loughborough University Ethical Advisory Committee, and participants provided written informed consent.

Procedure

At the beginning of the study, participants either attended a laboratory at Loughborough University or were visited by research staff at their place of work. During this meeting, participant's body mass (kg) and height (cm) were directly measured without shoes using electronic weighing scales (Tanita UK Ltd, Middlesex, UK) and a wall-mounted stadiometer (Seca UK, Birmingham, UK). Body mass index (BMI) was calculated as kg/m^2 , and general demographic information (age, sex, nature of employment, job title) was recorded. Participants were issued with an ActiGraph accelerometer and shown the correct wearing position. They were instructed to begin wearing the device upon waking up the following day. During the 7-day monitoring period, participants were requested to continue with their normal daily routine. Upon completion of the monitoring period, participants met with a researcher to return the accelerometer. During this meeting, they were asked to confirm if they had experienced a

From the School of Sport, Exercise and Health Sciences (Dr Clemes), Loughborough University, Loughborough, Leicestershire, United Kingdom; MoveLab (Dr O'Connell); Physical Activity and Exercise Research, Faculty of Medical Sciences, Newcastle University, Newcastle upon Tyne, United Kingdom; and Diabetes Research Centre (Dr Edwardson), University of Leicester, Leicester General Hospital, Leicester, Leicestershire, United Kingdom.

No external funding supported the work outlined in the article. The research was undertaken using existing funds and equipment within the School of Sport, Exercise and Health Sciences at Loughborough University.

The authors declare no conflicts of interest.

Address correspondence to: Stacy A. Clemes PhD, School of Sport, Exercise and Health Sciences, Loughborough University, Loughborough, Leicestershire LE11 3TU, UK (S.A.Clemes@lboro.ac.uk).

Copyright © 2014 by American College of Occupational and Environmental Medicine

DOI: 10.1097/JOM.0000000000000101

typical working week while wearing the device, and any days in which participants reported missing work through either illness or leave days were recorded.

Sedentary Behavior and Physical Activity Measurement and Data Processing

Participants wore an ActiGraph GT1M accelerometer (ActiGraph, Pensacola, FL) throughout waking hours for 7 consecutive days, except during water-based activities. The accelerometer was worn around the waist, above the midline of the thigh. The accelerometer was set to record data at 1-minute epochs. Accelerometer data were downloaded using ActiLife version 5 and processed using KinSoft version 3.3.75. Accelerometer data were considered valid if there were more than 600 minutes of monitoring per day (excluding continuous strings of zero counts for ≥ 60 minutes) recorded on at least 3 weekdays and 1 weekend day.²⁰ The widely used cut point of less than 100 counts/min (cpm) was used to estimate sedentary time (ie, estimated time spent sitting),²¹ whereas the Freedson cut points were used to estimate time spent in light-intensity activity (100 to 1951 cpm) (such as slow walking) and MVPA (such as brisk walking or jogging/running) (≥ 1952 cpm).²²

As preliminary analyses revealed that no significant differences occurred between the time spent in sedentary behavior and physical activity across Monday to Friday (data not shown), time spent in sedentary behavior, light-intensity activity, and MVPA was summarized for workdays (Monday to Friday in this sample) and nonworkdays (Saturday and Sunday). On workdays, time spent in each behavior was also summarized during working hours (9 AM to 4.59 PM) and during nonworking hours (before 9 AM and after 5 PM).

Statistical Analyses

Statistical analyses were conducted using IBM SPSS Statistics (IBM Corp., Armonk, NY) for Windows version 21. Time spent in sedentary behavior, light-intensity activity, and MVPA, along with the proportion of time spent in each behavior (accounting for accelerometer wear time), on workdays, nonworkdays, during working hours and nonworking hours on workdays was checked for normality using the one-sample Kolmogorov-Smirnov test, which showed that all data were not normally distributed. Nonparametric analyses were therefore undertaken, and the median and interquartile ranges (IQRs) were presented as descriptors throughout. To account for differences in accelerometer wear time during and outside working hours, comparisons were undertaken using the proportion of wear time spent in each behavior (sedentary, light-intensity activity, MVPA) as opposed to the absolute minute data. Specifically, the proportion of time spent in each behavior was compared between workdays and nonworkdays, and between working hours and nonworking hours on workdays using Wilcoxon-signed rank tests.

To address the secondary aim of this study, participants were grouped into tertiles on the basis of the proportion of time spent sedentary during working hours. Tertile 1 (lowest working-hours sedentary behavior) consisted of individuals who spent less than 68% of their working hours sedentary ($n = 55$). Tertile 2 (medium working-hours sedentary behavior) consisted of individuals who spent between 68% and 74% of their working hours sedentary ($n = 54$), and tertile 3 (highest working-hours sedentary behavior) consisted of individuals who were sedentary during working hours for equal to or above 75% of the time ($n = 61$). The three groups were compared in terms of the proportion of accelerometer wear time spent in sedentary behavior, light-intensity activity, and MVPA on nonworkdays and during nonworking hours on workdays using Kruskal-Wallis tests with Bonferroni-corrected post hoc comparisons. Age and BMI were also compared between the three groups using Kruskal-Wallis tests with Bonferroni-corrected post hoc comparisons. To further explore any links between sedentary behavior accumulated during and outside working hours, Spearman correla-

tions examined whether there were any associations between sedentary behavior measured during working hours and sedentary behavior accumulated on nonworkdays, and during nonworking hours on workdays. Statistical significance was set at $P < 0.05$ for all analyses unless otherwise stated.

To understand the pattern of sedentary behavior and physical activity occurring throughout the day, line graphs were constructed depicting the mean minutes per hour spent in sedentary behavior, light-intensity activity, and MVPA across the typical wear period (7 AM to 11.59 PM) for workdays and nonworkdays. The line graphs only contained data from valid days (> 10 hours) and hours (all 60 minutes) in which the accelerometer was worn by each participant. Separate graphs were created for the three tertiles for working-hours sedentary behavior described above in order for any differences in patterns between the groups to be identified.

RESULTS

Of the 210 participants who commenced the study, 170 (30% male; mean age, 40.1 ± 12.7 years; mean BMI, 24.5 ± 3.8 kg/m²) provided valid data and were included in the analyses. There were no significant differences between those who provided valid data and those who did not in terms of age, BMI, or sex proportion ($P > 0.05$). Males and females did not differ significantly in terms of the proportion of wear time spent in sedentary behavior and light-intensity physical activity during working and nonworking hours on workdays (all $P > 0.05$). Overall on workdays, males spent a significantly greater proportion of time and minutes in MVPA in comparison with females (median $\pm$ IQR, $4\% \pm 3\%$ vs $3\% \pm 3\%$; $P = 0.01$; 38 min/d vs 30 min/d; $P = 0.01$). There were no significant differences in the proportion of time spent in sedentary behavior, or in light-intensity activity and MVPA between males and females on nonworkdays (all $P > 0.05$; data not shown). Given the limited differences in the proportion of time spent in each behavior during and outside working hours between males and females, the analyses presented below focus on the sample as a whole.

Median accelerometer wear time was 874 ± 103 min/d on workdays and 767 ± 113 min/d on nonworkdays ($P < 0.001$); the sample provided valid accelerometer data (wear time ≥ 10 hrs/d) on 7 days per person (median value). Given the significant differences in wear time between the days (and between working hours and nonworking hours on workdays, Table 1), the proportion of wear time spent in each behavior (sedentary, light-intensity activity, and MVPA) was compared during and outside working hours as opposed to the absolute minutes. On workdays, participants spent a significantly greater proportion of time in sedentary behaviors, and significantly less time in light-intensity physical activity in comparison with nonworkdays (Table 1). There were no significant differences between workdays and nonworkdays in terms of the proportion of time spent in MVPA.

On workdays only, participants spent a greater proportion of time in sedentary behavior during working hours, and less time in light-intensity physical activity in comparison with nonworking hours (Table 1). Overall, sedentary behavior accumulated during working hours accounted for 57% of total daily sedentary time on workdays. There were no significant differences in the proportion of time spent in MVPA during working and nonworking hours on workdays.

When grouped into tertiles according to the proportion of working hours spent sedentary, significant differences in sedentary behavior and light-intensity physical activity were observed between the groups during nonworking hours (Table 2). Participants in the lowest tertile for sedentary behavior at work spent significantly less time in sedentary behavior and more time in light-intensity physical activity than those in the medium and high tertiles on nonworkdays (post hoc analyses, all $P < 0.01$). The three groups did not differ significantly in terms of the proportion of time spent in MVPA

TABLE 1. Sedentary Behavior and Physical Activity Measured During and Outside Working Hours in 170 Office Workers*

	All Days (Median ± IQR)			Workdays Only (Median ± IQR)		
	Workdays	Nonworkdays	Differences† P	During Working Hours	Nonworking Hours	Differences† P
Number of valid days‡	781	303		781	781	
Wear time, min/d	874 ± 103	767 ± 113	<0.001	477 ± 15	406 ± 79	<0.001
Wear time spent sedentary, %	68 ± 9	60 ± 14	<0.001	71 ± 12	63 ± 12	<0.001
Time in sedentary behavior, min/d	580 ± 101	460 ± 105		333 ± 61	254 ± 72	
Wear time spent in light PA, %	28 ± 9	36 ± 14	<0.001	25 ± 11	33 ± 10	<0.001
Time in light PA, min/d	246 ± 90	278 ± 126		117 ± 55	130 ± 48	
Wear time spent in MVPA, %	4 ± 3	4 ± 4	0.40	4 ± 4	3 ± 5	0.82
Time in MVPA, min/d	32 ± 26	28 ± 33		17 ± 17	13 ± 17	

*Data represent the median and interquartile ranges.

†Comparisons undertaken using Wilcoxon signed rank tests. As significant differences in accelerometer wear time were observed between workdays and nonworkdays, and between working hours and nonworking hours, comparisons were undertaken between the proportion of accelerometer wear time spent in each behavior. Minutes spent in each behavior are also included in the table for comparison purposes.

‡The number of valid days (wear time ≥ 10 hrs/d) included in the analyses.

IQR, interquartile ranges; MVPA, moderate-to-vigorous physical activity; PA, physical activity.

TABLE 2. Sedentary Behavior and Physical Activity Measured During and Outside Working Hours in Office Workers Grouped Into Tertiles According to the Proportion of Working Hours Spent Sedentary*

	All Days (Median ± IQR)							
	Workdays				Nonworkdays			
	Tertile 1 (n = 55)	Tertile 2 (n = 54)	Tertile 3 (n = 61)	Between-Group Differences† P	Tertile 1 (n = 55)	Tertile 2 (n = 54)	Tertile 3 (n = 61)	Between-Group Differences† P
Number of valid days‡	260	251	270		97	95	111	
Wear time, min/d	888 ± 112	884 ± 87	850 ± 77	0.02	775 ± 120	764 ± 84	744 ± 135	0.16
Wear time spent sedentary, %	59 ± 9	69 ± 5	72 ± 6	<0.001	54 ± 18	61 ± 11	64 ± 13	<0.001
Time in sedentary behavior, min/d	508 ± 102	594 ± 79	609 ± 76		427 ± 149	479 ± 114	468 ± 79	
Wear time spent in light PA, %	37 ± 8	28 ± 4	23 ± 7	<0.001	41 ± 15	36 ± 10	31 ± 12	<0.001
Time in light PA, min/d	325 ± 87	246 ± 41	198 ± 74		311 ± 106	274 ± 117	230 ± 104	
Wear time spent in MVPA, %	4 ± 4	3 ± 2	3 ± 3	0.21	4 ± 4	3 ± 4	4 ± 5	0.53
Time in MVPA, min/d	35 ± 36	30 ± 18	31 ± 26		28 ± 32	26 ± 33	30 ± 33	
	Workdays Only (Median ± IQR)							
	During Working Hours				Nonworking Hours			
	Tertile 1 (n = 55)	Tertile 2 (n = 54)	Tertile 3 (n = 61)	Between-Group Differences† P	Tertile 1 (n = 55)	Tertile 2 (n = 54)	Tertile 3 (n = 61)	Between-Group Differences† P
Number of valid days‡	260	251	270		260	251	270	
Wear time, min/d	478 ± 13	478 ± 13	474 ± 17	0.26	420 ± 86	418 ± 55	387 ± 64	<0.01
Wear time spent sedentary, %	60 ± 14	71 ± 3	78 ± 4	<0.001	60 ± 12	65 ± 10	66 ± 13	<0.001
Time in sedentary behavior, min/d	286 ± 68	335 ± 17	365 ± 26		247 ± 80	263 ± 64	243 ± 63	
Wear time spent in light PA, %	35 ± 12	25 ± 3	19 ± 5	<0.001	37 ± 10	32 ± 7	29 ± 10	<0.001
Time in light PA, min/d	163 ± 52	118 ± 19	88 ± 24		150 ± 56	128 ± 41	117 ± 53	
Wear time spent in MVPA, %	4 ± 4	4 ± 3	3 ± 3	<0.001	4 ± 5	3 ± 3	4 ± 5	0.14
Time in MVPA, min/d	20 ± 19	17 ± 13	13 ± 13		13 ± 25	13 ± 14	16 ± 18	

*Data represent the median and inter-quartile ranges.

†Between-group comparisons undertaken using Kruskal-Wallis tests with Bonferroni-corrected post hoc comparisons. To account for differences in accelerometer wear time between groups, comparisons were undertaken between the proportion of accelerometer wear time spent in each behavior. Minutes spent in each behavior are also included in the table for comparison purposes.

‡The number of valid days (wear time ≥ 10 hrs/d) included in the analyses for each tertile group.

IQR, interquartile ranges; MVPA, moderate-to-vigorous physical activity; PA, physical activity.

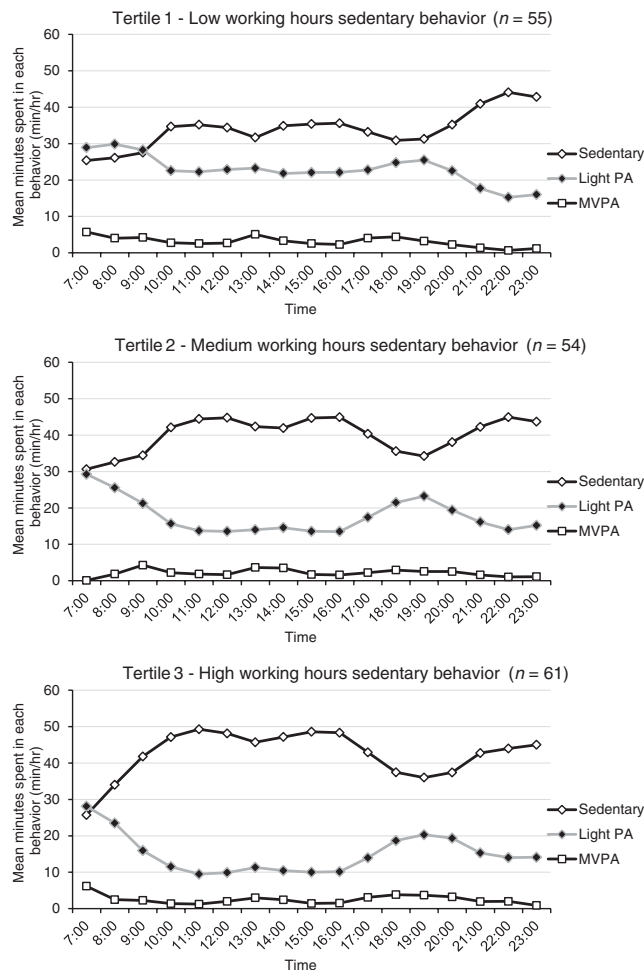


FIGURE 1. Minutes spent in sedentary behavior, light-intensity physical activity, and MVPA during each hour of the working day for participants grouped into tertiles on the basis of the proportion of time spent sedentary during working hours.

on nonworkdays (weekend days in this sample). Similarly, during nonworking hours on workdays, participants in the lowest tertile for sedentary behavior at work spent significantly less time in sedentary behavior and more time in light-intensity physical activity than those in the medium and high tertiles (post hoc analyses, all $P < 0.01$). Like nonworkdays, there were no significant differences between the groups in terms of the proportion of time spent in MVPA during nonworking hours on workdays (Table 2). There were no significant differences in BMI between participants in the three tertiles ($P > 0.05$). Nevertheless, participants in the lowest tertile for sedentary behavior at work were significantly older (46 ± 13 years) than those in the medium (38 ± 12 years) and high (36 ± 11 years) tertiles ($P < 0.01$).

For the sample as a whole, there were significant associations between the proportion of time spent sedentary during working hours and the proportion of time spent sedentary on nonworkdays ($r = 0.25$; $P < 0.001$), and during nonworking hours on workdays ($r = 0.36$; $P < 0.001$).

An hour-by-hour breakdown of the time (in minutes) spent in sedentary behavior, light-intensity activity, and MVPA on workdays and nonworkdays is shown in Figs. 1 and 2, respectively, for participants grouped into tertiles according to the proportion of

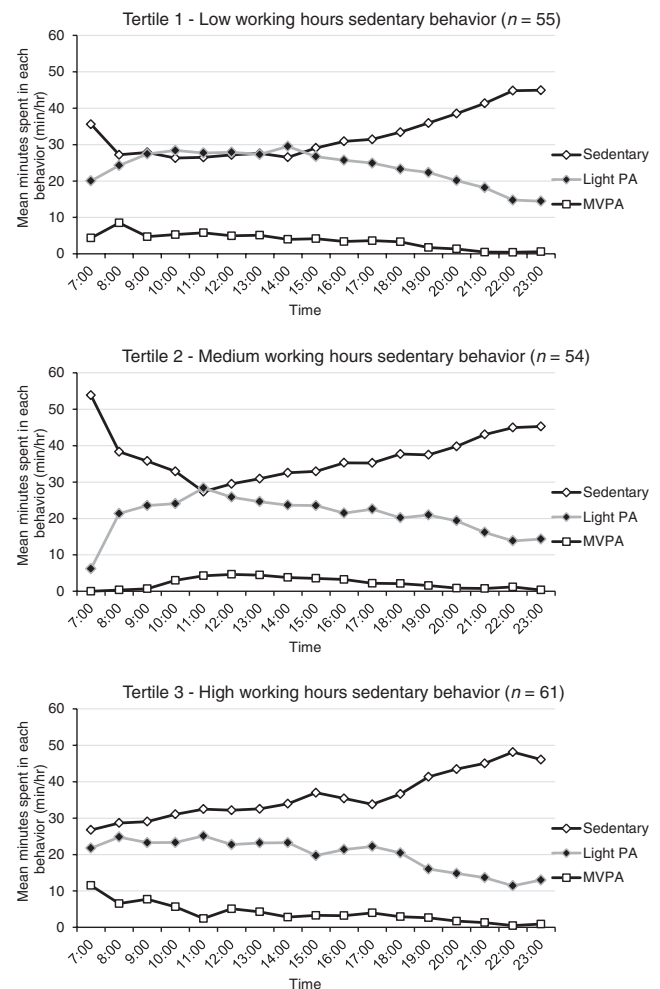


FIGURE 2. Minutes spent in sedentary behavior, light-intensity physical activity, and MVPA during each hour of the nonworking day for participants grouped into tertiles on the basis of the proportion of time spent sedentary during working hours.

time spent sedentary during working hours. On workdays, the three groups displayed a similar pattern in terms of the accumulation of sedentary behavior and light-intensity physical activity across the day; however, as to be expected on the basis of how the groups were defined (sedentary behavior during working hours), the difference between sedentary behavior and light-intensity activity over working hours becomes more pronounced across the groups. During working hours (9 AM to 4:59 PM), sedentary behavior was the most prominent behavior across all groups. All groups exhibited a small dip in this behavior around lunchtime followed by another dip immediately after working hours, which is then followed by a steady increase in sedentary behavior as the evening progresses. It is evident from Fig. 1 that on workdays, the pattern of light-intensity activity displays a mirror image of the pattern of sedentary behavior for all groups, suggesting that light-intensity activities offset sedentary behaviors. For all groups, MVPA displays a distinct pattern, showing small increases before working hours (7 AM to 8:59 AM), around lunchtime (1 PM to 1:59 PM) and after work into the early evening (5 PM to 7:59 PM).

The pattern of sedentary behavior and physical activity, accumulated hour by hour on nonworkdays (Fig. 2), differs from that seen for workdays (Fig. 1) for all groups. Through until mid-afternoon (8 AM to 3:59 PM), the proportion of sedentary behavior and

light-intensity activity is relatively equal for participants in the lowest tertile for working-hours sedentary behavior. From 4 PM onwards, sedentary behavior gradually increases throughout the evening as light-intensity activity decreases. A similar pattern can be observed in the medium tertile group; however, throughout the day sedentary behavior is the predominant behavior, with the steady increase in sedentary behavior and the decline in light-intensity activities starting earlier in the day (1 PM onwards). On nonworkdays, sedentary behavior is the most prominent behavior throughout the day for participants grouped in the highest tertile for working-hours sedentary behavior. The pattern of MVPA on nonworkdays seems to be similar across the groups, with MVPA being higher during the day, and decreasing from 7 PM onwards.

DISCUSSION

This study examined sedentary behavior and physical activity accumulated during and outside working hours in a sample of full-time office workers from the United Kingdom. On both workdays and nonworkdays, sedentary behavior was the most prevalent behavior exhibited by the sample, accounting for 68% and 60% of accelerometer wear time, respectively. On workdays, participants were highly sedentary during working hours, with 71% of working hours spent in sedentary behavior. Overall, sedentary behavior accumulated during working hours accounted for 57% of total daily sedentary time on workdays.

These findings add to the growing evidence highlighting the workplace as an important setting for the accumulation of high volumes of sedentary behavior.¹³ The proportion of working hours spent sedentary in the current sample is similar to that observed in Australian office workers, using objective measures.^{13–15} Given the workplace is the major contributor to total daily sedentary time on workdays, worksite interventions designed to reduce, or break up, sedentary behavior are urgently needed in UK office workers. Indeed, research in Australian and Swedish workers has started to investigate the effectiveness of sit-to-stand workstations for reducing sedentary time at work.^{23,24} If successful, the incorporation of sit-to-stand workstations in offices of sedentary workers within the UK workforce could be an effective strategy for reducing sedentary behavior during working hours.

It was observed in this study that sedentary behavior accumulated during working hours was positively associated with sedentary behavior measured on nonworkdays, and during nonworking hours on workdays. Furthermore, when split into tertiles according to the proportion of working hours spent sedentary, participants in the highest tertile for working-hours sedentary behavior spent a significantly greater proportion of time in sedentary behavior during nonworking hours on workdays and less time in light-intensity activity in comparison with participants in the lowest tertile for working-hours sedentary behavior. The same finding was also observed on nonworkdays. The observation that those who were most sedentary during working hours were also the most sedentary out of working hours is similar to that reported in Dutch¹⁶ and Australian¹⁴ workers. In this study, there were no significant differences between the groups in terms of the proportion of time spent in MVPA either during nonworking hours on workdays or on nonworkdays. This suggests that, in this sample, those who are sedentary for a large proportion of their working day do not compensate by increasing their physical activity levels outside working hours. This finding is in contrast to that reported by Chau et al,¹² who observed in Australian workers that individuals with jobs that mostly involve sitting were more likely to report being physically active during their leisure time than individuals in more active jobs. The differences in study findings may be attributable to the differences in lifestyles between these Australian and British samples, further highlighting the importance of understanding these lifestyle behaviors in different populations. Although participants in the three tertiles for working-hours sedentary behavior did not differ

in terms of job role, those in the lowest tertile were older than those in the medium and high tertiles, indicating that sedentary behavior levels and patterns may vary across age groups. This warrants further study in larger samples.

The finding that those who were most sedentary during working hours, were also the most sedentary during nonworking hours, coupled with the observation that there seems to be no compensatory increases in physical activity outside work, is a major concern. The “highest working-hours sedentary behavior” group spent more than 10 hours per day in sedentary behavior on workdays, suggesting that these individuals are at an increased risk of numerous chronic conditions associated with high volumes of sedentary behavior.⁸ In addition to an increased risk of chronic disease, evidence suggests that these individuals may also be at an increased risk of musculoskeletal disorders²⁵ and impaired work performance.¹⁵ On the basis of these findings, and others,^{14–16} it is suggested that worksite sedentary behavior interventions also target sedentary behavior outside working hours.

The hour-by-hour breakdown of time spent in each behavior for the three groups on workdays highlights working hours (9 AM to 4.59 PM) and the evening (8 PM onwards) as critical periods during the day when sedentary behavior is most prevalent. Although the overall pattern of behavior is similar on workdays across the three groups, the difference between light-intensity activity and sedentary behavior becomes more pronounced between the groups. Participants in the lowest tertile for working-hours sedentary behavior exhibited less time in sedentary behavior and a greater proportion of time in light-intensity activity in the hours before work, in comparison with the remaining groups. This difference could be down to differences in commuting behavior between the groups; however, as participants did not report their mode of transport to or from work in this study, this cannot be confirmed. For all groups on workdays (and nonworkdays), the pattern of light-intensity physical activity is the inverse to that of sedentary behavior, suggesting that light-intensity activities offset sedentary behaviors. Given the apparent strong link between sedentary behavior and light-intensity physical activity, workplace interventions promoting increases in light-intensity activity should be effective in reducing sedentary time. Given recent evidence suggesting that light-intensity physical activity is beneficial to health,²⁶ future worksite interventions targeting sedentary behavior should incorporate the promotion of light-intensity physical activity where feasible, such as encouraging the use of pooled printers/copiers, centrally placed water coolers, restricting e-mail and telephone contact for employees in the same building, etc. Emerging experimental evidence has shown that breaking up sedentary behavior every 20 minutes with 2 minutes of light walking significantly improves glucose and insulin regulation.²⁷ A strategy such as this could be implemented in future worksite interventions.

A small dip in sedentary behavior and increases in light-intensity activity and MVPA were observed around the lunch period on workdays, suggesting that this period could be a suitable time for encouraging longer breaks in sedentary behavior and increases in physical activity. Indeed, previous research has demonstrated the effectiveness of instructor-led lunchtime walking groups for promoting physical activity in sedentary workers.²⁸ In addition, recent research has shown that light-intensity physical activity during lunchtime was associated with reduced work performance impairment in office workers.¹⁵

This study provides novel information on how sedentary behavior and physical activity are accumulated during and outside working hours in a sample of office workers from the United Kingdom. The objective measurement of sedentary behavior and physical activity is a strength of this study as it likely overcomes the limitations of bias and recall, common with self-report measures. The study is not without its limitations however. Although the Acti-Graph accelerometer has been widely used as an objective measure

of sedentary behavior, this waist-worn device is not capable of distinguishing between standing and sitting/lying postures. Therefore, some periods of standing still may have been misclassified as sedentary behavior. Furthermore, in this study we applied the commonly used cut point of less than 100 cpm to estimate sedentary behavior. Despite its wide use, this cut point was not empirically derived, and recent contradictory evidence has questioned the validity of this particular cut point.^{29,30} For example, Kozey-Keadle et al²⁹ have suggested that a cut point of 150 cpm may be more accurate at defining sedentary time, although Hart et al³⁰ have reported that a cut point of less than 50 cpm may be more appropriate. Further research would benefit from the use of an inclinometer, as used elsewhere,¹⁷ which is capable of distinguishing between different postures. A further limitation of this study is that participants did not record their start and finish work times in a daily diary, and the working hours (9 AM to 5 PM) assigned in the this study were based on our knowledge of the standard working hours applied in the organizations in which participants were based. It is possible therefore that some of our participants may have been at work for longer or shorter periods than these assigned hours on some days of the study. Nevertheless, upon completion of the study, participants were asked to report whether they had had a typical week during the monitoring period, and any days where the participant had reported taking additional days off work through sickness or illness were removed ahead of the analyses. The cross-sectional design of the study prevents us from making conclusions about causality; it is therefore not possible to determine whether being sedentary at work leads to an individual being more sedentary out of working hours. Further longitudinal research is required to understand the long-term relationships between sedentary behavior accumulated during and outside working hours. Limited demographic information was collected from participants in this study; further research with larger samples should explore patterns of sedentary behavior occurring across different age groups, educational groups, and employment sectors, for example, to enhance the development of tailored interventions for reducing sedentary time.

CONCLUSIONS

This study extends our knowledge on the patterns of sedentary behavior and physical activity on workdays and nonworkdays in office workers living in the United Kingdom. The sample as a whole spent a large proportion of time in sedentary behavior on both workdays and nonworkdays. Of concern, it was observed in this study that those who are sedentary for a large proportion of their working hours also accumulate a high proportion of time in sedentary behavior during nonworking hours. There was no evidence to suggest that those with high volumes of sedentary behavior during working hours compensated for this by increasing their time in light-intensity activity or MVPA out of working hours. Given the high volume of sedentary behavior seen in this study, and others, workplace interventions are urgently needed to reduce sedentary time in adults to reduce the risk of numerous chronic diseases associated with sedentary behavior. Interventions should focus on reducing both workplace sedentary behavior and leisure-time sedentary behavior in sedentary office workers.

REFERENCES

1. Sedentary Behaviour Research Network. Standardized use of the terms "sedentary" and "sedentary behaviours". *Appl Physiol Nutr Metab*. 2012;37:540–542.
2. Hu FB, Li TY, Colditz GA, Willett WC, Manson JE. Television watching and other sedentary behaviors in relation to risk of obesity and type 2 diabetes mellitus in women. *JAMA*. 2003;289:1785–1791.
3. Brown WJ, Williams L, Ford JH, Ball K, Dobson AJ. Identifying the energy gap: magnitude and determinants of 5-year weight gain in midage women. *Obes Res*. 2005;13:1431–1441.
4. Gierach GL, Chang SC, Brinton LA, et al. Physical activity, sedentary behavior, and endometrial cancer risk in the NIH-AARP Diet and Health Study. *Int J Cancer*. 2009;124:2139–2147.
5. Lynch BM. Sedentary behavior and cancer: a systematic review of the literature and proposed biological mechanisms. *Cancer Epidemiol Biomarkers Prev*. 2010;19:2691–2709.
6. Edwardson CL, Gorely T, Davies MJ, et al. Association of sedentary behaviour with metabolic syndrome: a meta-analysis. *PLoS One*. 2012;7:e34916.
7. Katzmarzyk PT, Church TS, Craig CL, Bouchard C. Sitting time and mortality from all causes, cardiovascular disease, and cancer. *Med Sci Sports Exerc*. 2009;41:998–1005.
8. Wilmot EG, Edwardson CL, Achana FA, et al. Sedentary time in adults and the association with diabetes, cardiovascular disease and death: systematic review and meta-analysis. *Diabetologia*. 2012;55:2895–2905.
9. Hamilton MT, Hamilton DG, Zderic TW. Role of low energy expenditure and sitting in obesity, metabolic syndrome, type 2 diabetes, and cardiovascular disease. *Diabetes*. 2007;56:2655–2667.
10. Chau JY, der Ploeg HP, van Uffelen JG, et al. Are workplace interventions to reduce sitting effective? A systematic review. *Prev Med*. 2010;51:352–356.
11. Miller R, Brown W. Steps and sitting in a working population. *Int J Behav Med*. 2004;11:219–224.
12. Chau JY, van der Ploeg HP, Merom D, Chey T, Bauman AE. Cross-sectional associations between occupational and leisure-time sitting, physical activity and obesity in working adults. *Prev Med*. 2012;54:195–200.
13. Thorp AA, Healy GN, Winkler E, et al. Prolonged sedentary time and physical activity in workplace and non-work contexts: a cross-sectional study of office, customer service and call centre employees. *Int J Behav Nutr Phys Act*. 2012;9:128.
14. Parry S, Straker L. The contribution of office work to sedentary behaviour associated risk. *BMC Public Health*. 2013;13:296.
15. Brown HE, Ryde GC, Gilson ND, Burton NW, Brown WJ. Objectively measured sedentary behavior and physical activity in office employees: relationships with presenteeism. *J Occup Environ Med*. 2013;55:945–953.
16. Jans MP, Proper KI, Hildebrandt VH. Sedentary behavior in Dutch workers: differences between occupations and business sectors. *Am J Prev Med*. 2007;33:450–454.
17. Tigbe WW, Lean ME, Granat MH. A physically active occupation does not result in compensatory inactivity during out-of-work hours. *Prev Med*. 2011;53:48–52.
18. Stamatakis E, Hamer M, Dunstan DW. Screen-based entertainment time, all-cause mortality, and cardiovascular events: population-based study with ongoing mortality and hospital events follow-up. *J Am Coll Cardiol*. 2011;57:292–299.
19. Clemes SA, David BM, Zhao Y, Han X, Brown W. Validity of two self-report measures of sitting time. *J Phys Act Health*. 2012;9:533–539.
20. Matthews CE, Ainsworth BE, Thompson RW, Bassett DR Jr. Sources of variance in daily physical activity levels as measured by an accelerometer. *Med Sci Sports Exerc*. 2002;34:1376–1381.
21. Atkin AJ, Gorely T, Clemes SA, et al. Methods of measurement in epidemiology: sedentary behaviour. *Int J Epidemiol*. 2012;41:1460–1471.
22. Freedson PS, Melanson E, Sirard J. Calibration of the Computer Science and Applications, Inc. accelerometer. *Med Sci Sports Exerc*. 1998;30:777–781.
23. Healy GN, Eakin EG, Lamontagne AD, et al. Reducing sitting time in office workers: short-term efficacy of a multicomponent intervention. *Prev Med*. 2013;57:43–48.
24. Straker L, Abbott RA, Heiden M, Mathiassen SE, Toomingas A. Sit-stand desks in call centres: associations of use and ergonomics awareness with sedentary behavior. *Appl Ergon*. 2013;44:517–522.
25. Gerr F, Marcus M, Ensor C, et al. A prospective study of computer users: I. study design and incidence of musculoskeletal symptoms and disorders. *Am J Ind Med*. 2002;41:221–235.
26. Healy GN, Dunstan DW, Salmon J, et al. Objectively measured light-intensity physical activity is independently associated with 2-h plasma glucose. *Diabetes Care*. 2007;30:1384–1389.
27. Dunstan DW, Kingwell BA, Larsen R, et al. Breaking up prolonged sitting reduces postprandial glucose and insulin responses. *Diabetes Care*. 2012;35:976–983.
28. Chan CB, Ryan DA, Tudor-Locke C. Health benefits of a pedometer-based physical activity intervention in sedentary workers. *Prev Med*. 2004;39:1215–1222.
29. Kozey-Keadle S, Libertine A, Lyden K, Staudenmayer J, Freedson PS. Validation of wearable monitors for assessing sedentary behavior. *Med Sci Sports Exerc*. 2011;43:1561–1567.
30. Hart TL, McClain JJ, Tudor-Locke C. Controlled and free-living evaluation of objective measures of sedentary and active behaviors. *J Phys Act Health*. 2011;8:848–857.